

Lecture 5 Tutorial — Counting Theories & Naive Bayes

Q1. Bayes' Rule Components (Matching)

Match each term to the correct name in the lecture.

Given:

$$P(c \mid x) = \frac{P(c)P(x \mid c)}{P(x)}$$

Terms:

- A. $P(c)$
- B. $P(x \mid c)$
- C. $P(c \mid x)$
- D. $P(x)$

Names:

- 1. Prior probability
- 2. Likelihood
- 3. Posterior probability
- 4. Evidence factor (observation)

Q2. Likelihood Computation (Coin Models)

An experiment result is:

$$D = \text{"THHTTTHTTH"}$$

Consider two candidate models:

$$M_1 : p(H) = 0.5, p(T) = 0.5$$

$$M_2 : p(H) = 0.4, p(T) = 0.6$$

- 1. Count the number of H and T in D .
- 2. Compute $L_1 = p(D \mid M_1)$.
- 3. Compute $L_2 = p(D \mid M_2)$.
- 4. Select the model with higher likelihood (circle one): M_1 / M_2 .

Q3. Log-likelihood (Numerical Stability)

Using the same D , compute the log-likelihood difference:

$$\Delta = \log p(D \mid M_2) - \log p(D \mid M_1)$$

- 1. Write Δ in terms of $\log(0.4)$, $\log(0.6)$, and $\log(0.5)$.
 - 2. State whether $\Delta > 0$, $\Delta = 0$, or $\Delta < 0$ (no calculator required).
- (Use any log base; the sign must be consistent.)

Q4. Maximum Likelihood Estimation (MLE) for a Coin

For Bernoulli trials, let $\theta = p(H)$. Using “THHTTTHTTTH”:

1. Write the likelihood in the form $p(D | \theta) = \theta^k(1 - \theta)^{n-k}$.
2. Give the MLE $\hat{\theta}$ as a number.

(Answer only; no derivation required.)

Q5. Occam's Razor / Size Principle (Model Likelihood)

Let

$$D = \{16, 8, 2, 64\}.$$

Consider two hypotheses:

h_{two} : “powers of two under 100”

h_{even} : “even numbers below 100”

Assume each hypothesis defines a finite set and is uniform over that set. That is, if x is consistent with h , then

$$p(x | h) = \frac{1}{|h|}.$$

1. Compute $|h_{\text{two}}|$.
2. Compute $|h_{\text{even}}|$.
3. Compute the likelihood ratio:

$$\frac{p(D | h_{\text{two}})}{p(D | h_{\text{even}})}.$$

(Leave your answer as an exact fraction or a power expression.)

Q6. Posterior Update (Equal Priors)

Continue Q5. Assume equal priors:

$$p(h_{\text{two}}) = p(h_{\text{even}}) = 0.5$$

A new sample arrives:

$$x_{\text{new}} = 32$$

1. Compute the unnormalised posteriors:

$$p(h_{\text{two}} | D \cup \{32\}) \propto p(D \cup \{32\} | h_{\text{two}})p(h_{\text{two}})$$

$$p(h_{\text{even}} | D \cup \{32\}) \propto p(D \cup \{32\} | h_{\text{even}})p(h_{\text{even}})$$

2. Decide which posterior is larger (circle one): h_{two} / h_{even} .

Q7. Entropy, Cross Entropy, KL Divergence (Fill in + Compute)

The lecture defines the KL divergence (relative entropy) between P and Q as:

$$KL(P||Q) = \sum_{i=1}^{\ell} p_i \log \frac{p_i}{q_i}$$

Let

$$P = (0.5, 0.5), \quad Q = (0.9, 0.1).$$

1. Compute $KL(P||Q)$ (use natural log or base-2 log; state the base).
2. State whether $KL(P||Q)$ can ever be negative (Yes / No).

Q8. Zero-point Problem and Laplacian Correction (Short Calculation)

A discrete model has symbols $\ell = 3$ with alphabet $\{a, b, c\}$. In a training dataset, the counts for a class are:

$$N_a = 2, \quad N_b = 0, \quad N_c = 1$$

1. Compute the unsmoothed MLE probabilities $p(a)$, $p(b)$, $p(c)$.
2. Apply Laplacian correction (add-one smoothing) and compute the smoothed probabilities.

Q9. MAP vs MLE (Multiple Choice)

Select the most accurate statement.

- A. MLE selects the model with the highest posterior probability.
- B. MAP selects the model with the highest likelihood only.
- C. MAP maximises $p(c | x)$ and uses a term proportional to $p(c)p(x | c)$.
- D. MAP ignores the prior probability.

Q10. Naive Bayes Independence Assumption (Fill in)

Complete the Naive Bayes factorisation shown in the lecture:

If

$$x = (x_1, x_2, \dots, x_\ell),$$

then

$$p(x | c) = p(x_1 | c) p(x_2 | c) \cdots p(x_\ell | c).$$

This factorisation is based on the assumption that the features are _____ given the class c .

Q11. Naive Bayes Classification (Exam-style Computation)

Use the following training data (14 samples). Each row is:

(Outlook, Temperature, Humidity, Windy, Play)

Overcast	Hot	High	False	Yes
Overcast	Cool	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Rainy	Mild	Normal	False	Yes
Rainy	Mild	High	True	No
Sunny	Hot	High	True	No
Sunny	Hot	High	False	No
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Sunny	Mild	Normal	True	Yes

Classify the query:

$$x = (\text{Outlook}=\text{Overcast}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{Normal}, \text{Windy}=\text{True})$$

Part A (No smoothing)

1. Compute the unnormalised scores:

$$S_{\text{Yes}} = p(\text{Yes}) \prod_j p(x_j \mid \text{Yes})$$

$$S_{\text{No}} = p(\text{No}) \prod_j p(x_j \mid \text{No})$$

2. Which class is predicted by MAP? (Yes / No)

Part B (With Laplacian correction)

Apply add-one smoothing separately per feature within each class with category counts:

- Outlook: 3 categories (Sunny, Overcast, Rainy)
- Temperature: 3 categories (Hot, Mild, Cool)
- Humidity: 2 categories (High, Normal)
- Windy: 2 categories (True, False)

1. Recompute the two unnormalised scores S_{Yes} and S_{No} using the smoothed conditionals.
2. Which class is predicted now? (Yes / No)

Q12. Bayesian Network (Multiple Choice)

A Bayesian Network is described as using a Directed Acyclic Graph (DAG) to represent dependencies and factorise a joint distribution.

Which statement is most consistent with this description?

- A. It replaces probability with deterministic rules.
- B. It encodes feature dependencies via a DAG and expresses a joint distribution as a product of conditional distributions.
- C. It assumes all features are independent without exception.
- D. It can only be used for coin and dice experiments.